

Paper-by-Paper Fit and Gap Matrix

Comparison against the MoltBook project focus: thread-level sentiment homeostasis, parent-child sentiment alignment, author sentiment consistency, interaction structure, and sentiment-method robustness.

Paper Link	Fit	What Matches Our Research	Gap Our Project Can Fill
The Rise of AI Agent Communities arXiv:2602.12634	Very high	Same platform; discourse, topics, broad sentiment tone, neutral writing, positivity in community contexts, sparse interaction network, hubs, low reciprocity.	Does not deeply model parent-child sentiment alignment, thread-phase trajectories, neutral attractor behavior, negative de-escalation, author sentiment consistency, or scorer robustness.
Let There Be Claws arXiv:2602.20044	High	Strong match for RQ1: directed comment graphs, reciprocity, community structure, centrality, attention inequality, hub-authority roles, memory and identity topics.	Mostly structural. It does not explain how sentiment behaves inside reply chains or whether hub agents stabilize discourse tonally.
"Humans welcome to observe" arXiv:2602.10127	High	Same platform; large-scale empirical analysis; topic taxonomy; toxicity and risk; temporal development; centralization and platform-level safeguards.	Risk and toxicity framing dominates. It does not focus on ordinary sentiment distribution, parent-child alignment, or rule-based sentiment robustness.
Agents in the Wild arXiv:2602.13284	High	Same platform; safety and society framing; low reciprocity; shallow interaction; pro-human vs anti-human sentiment; critique of sociality claims.	Main object is safety and structural hollowness, not sentiment homeostasis or thread-level affective regulation.
Does Socialization Emerge in AI Agent Society? arXiv:2602.14299	High	Same platform; semantic stabilization, individual inertia, weak mutual influence, lack of shared social memory. Very relevant to author-level sentiment consistency.	Studies semantic and socialization dynamics rather than sentiment-class transitions or reply-level affective alignment.
Benchmarking Emergent Coordination in Large-Scale LLM Populations arXiv:2603.03555	Medium	MoltBook archive; role specialization, information diffusion, core-periphery structure, coordination overhead.	Coordination-focused rather than sentiment-focused. It does not test ordinary sentiment transitions or method robustness.
Harm in AI-Driven Societies arXiv:2601.01090	High method fit	Strong methodological parallel: models posts as stimuli and comments as responses; studies whether harmful exposure predicts harmful replies.	Different platform and toxicity-only outcome. It does not study MoltBook or general positive/neutral/negative sentiment dynamics.

Paper Link	Fit	What Matches Our Research	Gap Our Project Can Fill
Collective Behaviors and Social Dynamics in LLM Agents arXiv:2602.03775 / EACL 2026	Medium-high	AI-agent social network; homophily, social influence, toxicity, ideological leaning, polarization, longitudinal dynamics.	Chirper.ai rather than MoltBook. Broader social dynamics, not thread-level sentiment homeostasis.
Characterizing LLM-driven Social Network: The Chirper.ai Case arXiv:2504.10286	Medium	AI-agent social network compared with human social network; posting behavior, abusive content, and social structure.	Different platform. It does not address MoltBook-specific comment threads, neutral attractor behavior, or scorer sensitivity.
Unveiling Collective Behaviors of LLM-based Autonomous Agents Data and Information Management, 2026	Medium	Chirper-based social network analysis; centralized interaction, power-law degree patterns, homophily, machine behavior framing.	Does not address MoltBook, sentiment alignment, or scorer robustness.
Generative Agents arXiv:2304.03442	Background	Foundational agent society architecture; memory, reflection, planning, emergent social behavior.	Simulation rather than observed platform data. It does not compete with your empirical MoltBook analysis.